

2024, Mar. 2024 Release LOC:Acute Adult

Inflammatory Bowel Disease

Utilization Benchmarks: Multiple options available, see table below for options

Overview

Select Day

Initial review ^(1, 2)Episode Day 1 ⁽¹⁾Episode Day 2 ⁽³⁰⁾Episode Day 3 ⁽⁴⁶⁾

Episode Day 4

Episode Day 5

Discharge Screens ⁽⁵⁰⁾

Utilization Benchmarks

Notes

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Overview

Instruction:

This subset is for patients with Crohn's disease (CD) or ulcerative colitis (UC). It is intended for the patient who has a history of inflammatory bowel disease (IBD) presenting with an acute flare, bowel obstruction, fistula, ileus, or megacolon. For patients with either Crohn's disease or ulcerative colitis who present with sepsis or systemic inflammatory response syndrome (SIRS), use the Infection: Sepsis subset. For toxic megacolon requiring surgical intervention use the General Surgical subset. For an intra-abdominal abscess use the Infection: GI/GYN subset.

Introduction:

IBD is comprised of two major diseases, CD and UC. Each is characterized by chronic inflammation of the bowel with periods of remission and exacerbation. The etiology is unknown, but genetic factors, infectious agents, environmental triggers, and impaired immune regulation have been proposed as causative factors.

Evaluation and Treatment:

CD is characterized by transmural inflammation (full-thickness) of the intestine affecting any part of the gastrointestinal tract from mouth to anus, but most commonly affecting the ileum or colon. Clinical presentation includes chronic diarrhea, abdominal pain (often worse postprandially), weight loss, and fever. CD can present with extraintestinal manifestations, such as large joint arthritis, uveitis, iritis, episcleritis, erythema nodosum and pyoderma gangrenosum. A clinical diagnosis is determined by endoscopic biopsy, radiology findings, history, and evidence of chronic intestinal inflammation. Chronic intestinal inflammation can lead to intestinal complications such as strictures, fistulas, and abscesses. Surgical interventions including bowel resection, stricturoplasty or drainage of abscess can be indicated in up to two thirds of CD patients in their lifetime. Refractory disease, perforation, persisting or recurrent obstruction, abscess not responsive to percutaneous drainage, intractable hemorrhage, dysplasia or cancer are all indications for surgical resection. Early surgical consultation should be considered in all patients hospitalized with IBD. UC is characterized by chronic inflammation of the colonic mucosa of the large bowel. Common symptoms of UC may include frequent bloody diarrhea, which is often accompanied by rectal urgency and tenesmus. Extraintestinal manifestations may include axial and peripheral arthropathies, uveitis, sclerosing cholangitis, erythema nodosum and pyoderma gangrenosum. The diagnosis is confirmed by endoscopic biopsy. Possible complications include bleeding, toxic or fulminant colitis, megacolon, and increased risk of colorectal cancer.

Typical management of moderate to severe inflammatory bowel disease consists of corticosteroids, immunosuppressants (e.g., azathioprine, 6-mercaptopurine, methotrexate, cyclosporin, tacrolimus), or biologics such as infliximab (Remicade), adalimumab (Humira), certolizumab pegol (Cimzia), Golimumab (Simponi), natalizumab (Tysabri), and vedolizumab (Entyvio). The goals of treatment include: induce mucosal healing, promote and maintain

remission of symptoms, improve quality of life, minimize the need for corticosteroids, and reduce the risk of megacolon or gastrointestinal cancer.

In a patient with severe UC not responding to maximum immunosuppressive therapy, cytomegalovirus (CMV) superinfection should be considered. The diagnosis of CMV is confirmed by endoscopic biopsy. CMV colitis is treated with intravenous ganciclovir and may require surgery.

Rates of *Clostridioides difficile* (*C. difficile*) infection have increased globally in both patients with IBD and non-IBD patients. Unlike patients without IBD, these patients do not need to have the risk factors of recent hospitalization or antibiotic use, to have a high suspicion or get tested for *C. difficile*. *C. difficile* should be ruled out as a superimposed infection by stool examination. It is recommended that patients with IBD and *C. difficile* be treated with vancomycin instead of metronidazole.

Another complication of IBD that requires hospitalization is megacolon. Megacolon is diagnosed by abdominal x-ray, ultrasound, or CT scan. These patients are treated with steroids and intravenous fluid. Due to high risk of perforation and surgical intervention these patients are to remain NPO. Toxic megacolon is a medical emergency requiring intervention at the critical level of care and likely surgery.

References: (Kaur et al., Clinical Gastroenterology and Hepatology 2020, 18: 1346-55; National Institute for Health and Clinical Excellence (NICE), NICE Clinical guideline 129. Crohns Disease Management. 2019.; Rubin et al., American Journal of Gastroenterology 2019, 114: 384-413; Gajendran et al., Disease-a-month 2018, 64: 20-57; Lichtenstein et al., Am J Gastroenterol 2018, 113: 481-517; Veauthier and Hornecker, American family physician 2018, 98: 661-9)

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽³⁾● Inflammatory bowel disease, acute flare, **Both:**– Failed outpatient medication therapy (includes PO) ^(4, 5)● Ulcerative colitis and, ≥ **One:** ⁽⁶⁾

– 4-6 stools within the last 24h

– Intermittent hematochezia ⁽⁷⁾● **ACUTE, ≥ One:** ⁽⁸⁾● Inflammatory bowel disease, acute flare, **Both:**– Failed outpatient medication therapy (includes PO) ^(4, 5)● Finding, **One:**● Ulcerative colitis and, ≥ **One:** ⁽⁶⁾

– ≥ 7 stools within the last 24h

– Persistent hematochezia ⁽⁷⁾● Crohn's disease and, ≥ **One:** ⁽⁹⁾

– Abdominal pain or tenderness

– Persistent diarrhea

– Persistent vomiting

– Temperature ≥ 100.4°F(38.0°C) ⁽¹⁰⁾● Fistula, ≥ **One:** ^(11, 12, 13)

– Enterocutaneous

● Enteroenteric and, **One:**– Confirmed by CT scan ⁽¹⁴⁾● Suspected, ≥ **One:**

– Abdominal pain

– Palpable mass

– Ileus confirmed by x-ray or CT ⁽¹⁵⁾– Megacolon confirmed by x-ray, ultrasound, or CT ⁽¹⁶⁾– Small bowel obstruction confirmed by x-ray or CT ⁽¹⁷⁾● **CRITICAL, ≥ One:** ⁽¹⁸⁾● Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ **One:** ⁽¹⁹⁾

– MAP < 65 mmHg

– Systolic blood pressure < 90 mmHg and < baseline

– Hemodynamic monitoring, invasive ⁽²⁰⁾

– Mechanical ventilation

● NIPPV and, ≥ **One:** ⁽²¹⁾– Arterial Po₂ ≤ 39 mmHg(5.2 kPa)● Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**

– COPD and arterial or venous pH ≤ 7.24

– No hx of COPD

– Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 1, One:** ⁽¹⁾

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽³⁾● Inflammatory bowel disease, acute flare, **Both:**– Failed outpatient medication therapy (includes PO) ^(4, 5)● Ulcerative colitis and, **Both:** ⁽⁴⁾● Findings, ≥ **One:** ⁽⁶⁾

- 4-6 stools within the last 24h
 - Intermittent hematochezia ⁽⁷⁾
- Intervention, **Both:** ^(22, 23)
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - IV fluid ⁽²⁵⁾
- ACUTE, ≥ **One:** ⁽⁸⁾
 - Surgical intervention required (**use General Surgical subset**)
- Inflammatory bowel disease, acute flare, **All:**
 - Failed outpatient medication therapy (includes PO) ^(4, 5)
- Finding, **One:**
 - Ulcerative colitis and, ≥ **One:** ⁽⁶⁾
 - ≥ 7 stools within the last 24h
 - Persistent hematochezia ⁽⁷⁾
 - Crohn's disease and, ≥ **One:** ^(9, 26)
 - Abdominal pain or tenderness
 - Persistent diarrhea
 - Persistent vomiting
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁰⁾
- Intervention, **Both:** ^(22, 23)
 - IV fluid ⁽²⁵⁾
 - Immunosuppressant (includes PO) ⁽²⁴⁾
- Fistula, **Both:** ^(11, 12, 13)
 - Finding, ≥ **One:**
 - Enterocutaneous
 - Enteroenteric and, **One:**
 - Confirmed by CT
 - Suspected, ≥ **One:**
 - Abdominal pain
 - Palpable mass
- Intervention, ≥ **One:** ⁽²³⁾
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
- Ileus confirmed by x-ray or CT and, **All:** ⁽¹⁵⁾
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - IV fluid ⁽²⁵⁾
 - NPO
- Megacolon confirmed by x-ray, ultrasound, or CT and, **All:** ⁽¹⁶⁾
 - Corticosteroid (includes PO) ⁽²⁴⁾
 - IV fluid ⁽²⁵⁾
 - NPO
- Small bowel obstruction confirmed by x-ray or CT and, **All:** ^(17, 27, 28)
 - IV fluid ⁽²⁵⁾
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - Nasogastric tube to suction ⁽²⁹⁾
- CRITICAL, ≥ **One:** ⁽¹⁸⁾
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ **One:** ⁽¹⁹⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Hemodynamic monitoring, invasive ⁽²⁰⁾
 - Mechanical ventilation
 - NIPPV and, ≥ **One:** ⁽²¹⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)

- Arterial or venous $P_{CO_2} \geq 60$ mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2, One:** ⁽³⁰⁾
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **Both:**
 - Frequency of stools or hematochezia improving
 - Tolerating PO or nutritional route established
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
 - Ulcerative colitis, acute flare and, **Both:** ⁽⁴⁾
 - Findings, $\geq$ **One:** ⁽⁶⁾
 - 4-6 stools within the last 24h
 - Intermittent hematochezia ⁽⁷⁾
 - Intervention, **Both:** ⁽²³⁾
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - Hydration or nutrition, $\geq$ **One:**
 - IV fluid ⁽²⁵⁾
 - Advancing diet as tolerated ⁽³¹⁾
 - **ACUTE, One:**
 - **Early responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³²⁾
 - Afebrile
 - Pain controlled or manageable ⁽³³⁾
 - Tolerating PO or nutritional route established ^(34, 35)
 - No complication or active comorbidity relevant to this episode of care ⁽³⁶⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - Surgical intervention required (**use General Surgical subset**)
 - Inflammatory bowel disease, acute flare and, **All:**
 - Finding, $\geq$ **One:**
 - Abdominal pain or tenderness
 - Persistent diarrhea or hematochezia
 - Persistent vomiting
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ ⁽¹⁰⁾
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - Hydration or nutrition, $\geq$ **One:**
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - **Fistula, $\geq$ One:** ^(11, 12, 13)
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - **Ileus, One:**
 - Unresolved and, **All:**
 - NPO
 - Hydration or nutrition, $\geq$ **One:**
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - Resolved and $\geq$ **One:**
 - Initiating PO ≤ 24 hours
 - Tube feeding ≤ 2 days
 - **Megacolon and, All:** ⁽¹⁶⁾
 - NPO
 - Corticosteroid (includes PO)

- Hydration or nutrition, ≥ **One**:
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - Small bowel obstruction, **One**: ^(17, 28)
 - Unresolved and, **All**:
 - Hydration or nutrition, ≥ **One**:
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - Nasogastric tube to suction ⁽²⁹⁾
 - Resolved and initiating PO or tube feeding ≤ 2d
 - Blood glucose ≤ 24h, **All**: ⁽³⁷⁾
 - Finding, ≥ **One**:
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(37, 38)
 - *C. difficile* infection, actual or suspected and, **Both**: ⁽³⁹⁾
 - Anti-infective (includes PO) ≤ 5d ^(40, 41)
 - Inadequate oral intake ⁽⁴²⁾
 - Post Critical care ≤ 24h
- **INTERMEDIATE**, ≥ **One**: ⁽⁴³⁾
 - NIPPV ⁽²¹⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽⁴⁴⁾
- **CRITICAL**, ≥ **One**:
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ **One**: ⁽¹⁹⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Hemodynamic monitoring, invasive ⁽²⁰⁾
 - Mechanical ventilation
 - NIPPV and, ≥ **One**: ⁽²¹⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁴⁵⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 3, One**: ⁽⁴⁶⁾
 - **OBSERVATION, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **Both**:
 - Frequency of stools or hematochezia improving
 - Tolerating PO or nutritional route established
 - **Non-responder**, not clinically stable for discharge and exceeds initial 48h Observation for this condition, **One**:
 - For ulcerative colitis use Episode Day 3 at a higher level of care
 - For a condition other than ulcerative colitis use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽³²⁾
 - Afebrile
 - Pain controlled or manageable ⁽³³⁾
 - Tolerating PO or nutritional route established ^(34, 35)
 - **Complication or comorbidity, One**:
 - No complication or active comorbidity relevant to this episode of care ⁽³⁶⁾

- Complication or comorbidity and clinically stable for discharge, ≥ **One**:
 - Blood glucose within acceptable limits ⁽⁴⁷⁾
 - *C. difficile* infection, symptoms resolving, and tolerating PO ⁽⁴⁸⁾
 - Outpatient fistula management regimen established ⁽⁴⁹⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Surgical intervention required (**use General Surgical subset**)
- Inflammatory bowel disease, acute flare and, **All**:
 - Finding, ≥ **One**:
 - Abdominal pain or tenderness
 - Persistent diarrhea or hematochezia
 - Persistent vomiting
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁰⁾
 - Immunosuppressant (includes PO) ⁽²⁴⁾
- Hydration or nutrition, ≥ **One**:
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - Advancing diet as tolerated ⁽³¹⁾
- **Fistula, One**: ^(11, 12, 13)
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
- **Ileus, One**:
 - Unresolved and, **All**:
 - NPO
 - Hydration or nutrition, **One**:
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - Resolved and initiating PO or tube feeding ≤ 2d
- Megacolon resolving on imaging and, **Both**: ⁽¹⁶⁾
 - Corticosteroid (includes PO)
- Hydration or nutrition, ≥ **One**:
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - Advancing diet as tolerated ⁽³¹⁾
- Small bowel obstruction, **One**: ^(17, 28)
 - Unresolved and, **All**:
 - Hydration or nutrition, **One**:
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - Nasogastric tube to suction ⁽²⁹⁾
 - Resolved and initiating PO or tube feeding ≤ 2d
- Blood glucose ≤ 24h, **All**: ⁽³⁷⁾
 - Finding, ≥ **One**:
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(37, 38)
- *C. difficile* infection, actual or suspected and, **Both**: ⁽³⁹⁾
 - Anti-infective (includes PO) ≤ 5d ^(40, 41)
 - Inadequate oral intake ⁽⁴²⁾
- Post Critical care ≤ 24h

● **INTERMEDIATE, ≥ One:**

- NIPPV ⁽²¹⁾
- Oxygen $\geq 40\%(0.40)$, $\leq 2d$ ⁽⁴⁴⁾
- **CRITICAL, $\geq$ One:**
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation $\leq 24h$ and, $\geq$ **One:** ⁽¹⁹⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Hemodynamic monitoring, invasive ⁽²⁰⁾
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽²¹⁾
 - $FiO_2 > 50\%(0.50)$
 - Respiratory intervention every 1-2h ⁽⁴⁵⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 4, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³²⁾
 - Afebrile
 - Pain controlled or manageable ⁽³³⁾
 - Tolerating PO or nutritional route established ^(34, 35)
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽³⁶⁾
 - **Complication or comorbidity and clinically stable for discharge, $\geq$ One:**
 - Blood glucose within acceptable limits ⁽⁴⁷⁾
 - *C. difficile* infection, symptoms resolving, and tolerating PO ⁽⁴⁸⁾
 - Outpatient fistula management regimen established ⁽⁴⁹⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - Surgical intervention required (**use General Surgical subset**)
 - **Inflammatory bowel disease, acute flare and, Both:**
 - **Finding, $\geq$ One:**
 - Abdominal pain or tenderness
 - Persistent diarrhea or hematochezia
 - Persistent vomiting
 - Temperature $\geq 100.4^\circ F(38.0^\circ C)$ ⁽¹⁰⁾
 - **Intervention, Both:**
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - **Nutrition, $\geq$ One:**
 - TPN or PPN
 - Advancing diet as tolerated ⁽³¹⁾
 - **Fistula with continued output and, $\geq$ One:** ^(11, 12, 13)
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - Fistula management and teaching $\leq 24h$ ⁽⁴⁹⁾
 - **Ileus, One:**
 - **Unresolved and, All:**
 - NPO
 - **Hydration or nutrition, One:**
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - Resolved and initiating PO or tube feeding $\leq 2d$
 - **Megacolon resolving on imaging and, Both:** ⁽¹⁶⁾
 - Corticosteroid (includes PO)
 - **Nutrition, $\geq$ One:**

- TPN or PPN
 - Advancing diet as tolerated $\leq 2d$ ⁽³¹⁾
- Small bowel obstruction, **One:** ^(17, 28)
 - Unresolved and, **All:**
 - Hydration or nutrition, $\geq$ **One:**
 - IV fluid ⁽²⁵⁾
 - TPN or PPN
 - Immunosuppressant (includes PO) ⁽²⁴⁾
 - Nasogastric tube to suction ⁽²⁹⁾
 - Resolved and initiating PO or tube feeding $\leq 2d$
- Blood glucose $\leq 24h$, **All:** ⁽³⁷⁾
 - Finding, $\geq$ **One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(37, 38)
- *C. difficile* infection, actual or suspected and, **Both:** ⁽³⁹⁾
 - Anti-infective (includes PO) $\leq 5d$ ^(40, 41)
 - Inadequate oral intake ⁽⁴²⁾
 - Post Critical care $\leq 24h$
- **INTERMEDIATE, $\geq$ One:**
 - NIPPV ⁽²¹⁾
 - Oxygen $\geq 40\%(0.40)$, $\leq 2d$ ⁽⁴⁴⁾
- **CRITICAL, $\geq$ One:**
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation $\leq 24h$ and, $\geq$ **One:** ⁽¹⁹⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Hemodynamic monitoring, invasive ⁽²⁰⁾
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽²¹⁾
 - $FiO_2 > 50\%(0.50)$
 - Respiratory intervention every 1-2h ⁽⁴⁵⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 5, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³²⁾
 - Afebrile
 - Pain controlled or manageable ⁽³³⁾
 - Tolerating PO or nutritional route established ^(34, 35)
 - Complication or comorbidity, **One:**
 - No complication or active comorbidity relevant to this episode of care ⁽³⁶⁾
 - Complication or comorbidity and clinically stable for discharge, $\geq$ **One:**
 - Blood glucose within acceptable limits ⁽⁴⁷⁾
 - *C. difficile* infection, symptoms resolving, and tolerating PO ⁽⁴⁸⁾
 - Outpatient fistula management regimen established ⁽⁴⁹⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **INTERMEDIATE, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**

- Use Extended Stay subset
- Refer for secondary review

● **CRITICAL, One:**

- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - Use Extended Stay subset
 - Refer for secondary review
-

Discharge, Level of Care, One: ⁽⁵⁰⁾

● **HOME, Both:**

- Level of care appropriateness, **Both:**
 - Home environment safe and accessible ⁽⁵¹⁾
 - Patient or caregiver demonstrates ability to manage care
- Inflammatory bowel disease discharge planning, **All:** ⁽⁵²⁾
 - Provide comprehensive written discharge and teaching instructions ⁽⁵³⁾
- Education on the importance of, **All:**
 - Compliance with follow-up appointments
 - Medication compliance
 - Tapering steroids and problems with stopping abruptly
 - Understanding flare triggers, signs and symptoms of flare, and interventions to reduce symptoms
 - When to seek medical attention
 - Smoking cessation
- Medication reconciliation ⁽⁵⁴⁾
- Follow-up care planned ⁽⁵⁵⁾
- Identify and address transportation needs

● **HOME CARE, Both:**

- Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽⁵¹⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(56, 57)
- Skilled services, $\geq$ **One:** ⁽⁵⁸⁾
 - Skilled nursing ⁽⁵⁹⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁶⁰⁾
- Inflammatory bowel disease discharge planning, **All:** ⁽⁵²⁾
 - Provide comprehensive written discharge and teaching instructions ⁽⁵³⁾
- Education on the importance of, **All:**
 - Compliance with follow-up appointments
 - Medication compliance
 - Tapering steroids and problems with stopping abruptly
 - Understanding flare triggers, signs and symptoms of flare, and interventions to reduce symptoms
 - When to seek medical attention
 - Smoking cessation
- Medication reconciliation ⁽⁵⁴⁾
- Follow-up care planned ⁽⁵⁵⁾
- Identify and address transportation needs

● **BEHAVIORAL HEALTH, Both:**

- Level of care appropriateness, **One:**
 - Support systems available ⁽⁶¹⁾
- Skilled treatment, $\geq$ **One:** ⁽⁶²⁾
 - Clinical assessment ⁽⁶²⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁶³⁾

● SKILLED MEDICAL OR THERAPY, Both:**● Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 1x/wk$
- Treatment precluded in a lower level

● Services, $\geq$ One:

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁵⁹⁾

● Therapy, All:

- Goal directed therapy with rehabilitation potential ^(64, 65)
- Functional impairments requiring at least supervision ⁽⁶⁶⁾
- Able to tolerate 1h/d of skilled therapy $\geq 5d/wk$

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽⁵⁴⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● SUBACUTE MEDICAL OR THERAPY, Both:**● Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 2x/wk$
- Treatment precluded in a lower level

● Services, $\geq$ One:

- Medical and skilled nursing interventions or assessment 4h/24h ⁽⁵⁹⁾

● Therapy, All:

- Goal directed therapy with rehabilitation potential ^(64, 65)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁶⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 1 therapy discipline

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽⁵⁴⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● SUBACUTE REHAB, Both:**● Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
- Treatment precluded in a lower level

● Services, All:

- Goal directed therapy with rehabilitation potential ^(64, 65)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁶⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 2 therapy disciplines

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽⁵⁴⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● ACUTE REHAB, Both:**● Level of care appropriateness, Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$

● Services, All:

- Goal directed therapy with rehabilitation potential ^(64, 65)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁶⁾
- Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁶⁷⁾

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽⁵⁴⁾
- Obtain and complete forms for facility

- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **LONG TERM ACUTE CARE, Both:**

- **Level of care appropriateness, Both:**
 - Treatment precluded in a lower level
- **In lieu of acute or continued hospitalization or failed lower level of care, ≥ One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁶⁸⁾
 - Ventilator dependent ≥ 6h/d and weaning planned ^(69, 70)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁷¹⁾
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁵⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
385 INFLAMMATORY BOWEL DISEASE WITH MCC	n/a	5	CMS GMLOS
386 INFLAMMATORY BOWEL DISEASE WITH CC	n/a	3.4	CMS GMLOS
387 INFLAMMATORY BOWEL DISEASE WITHOUT CC/MCC	n/a	2.6	CMS GMLOS
388 GASTROINTESTINAL OBSTRUCTION WITH MCC	n/a	4.6	CMS GMLOS
389 GASTROINTESTINAL OBSTRUCTION WITH CC	n/a	3.1	CMS GMLOS
390 GASTROINTESTINAL OBSTRUCTION WITHOUT CC/MCC	n/a	2.3	CMS GMLOS
CROHN’S DISEASE	19%	3.8	InterQual
ULCERATIVE COLITIS	19%	4.3	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

When patients experience an acute flare of inflammatory bowel disease, inducing remission is often achieved by the administration of high dose oral corticosteroids. Dosage includes 40-60 mg/day or 1 mg/kg/day of prednisone or its equivalent. Hospitalization for parenteral corticosteroids is indicated when a patient fails to respond or is unable to tolerate oral corticosteroids (Rubin et al., American Journal of Gastroenterology 2019, 114: 384-413; Lichtenstein et al., Am J Gastroenterol 2018, 113: 481-517).

5:

Instruction: These criteria can be applied when the patient has received more intensive treatment beyond their usual treatment regimen. The location (e.g., medical practitioner's office, outpatient clinic, or trial of home services) in which the treatment occurs does not impact the application of these criteria. Medications taken at the patient's usual dosage and frequency for this condition do not apply.

6:

The Truelove-Witt Severity Index may be used to determine severity of ulcerative colitis. Severe ulcerative colitis is demonstrated by more than 6 stools daily (Rubin et al., American Journal of Gastroenterology 2019, 114: 384-413; Dulai and Jairath, Ther Adv Chronic Dis 2018, 9: 65-72; Sinagra et al., J Biol Regul Homeost Agents 2018, 32: 415-23). Patients with mild to moderate UC may be effectively treated in the outpatient setting.

7:

Hematochezia is defined as maroon or bright red blood passed from the rectum and usually signifies a lower gastrointestinal source. In these criteria, hematochezia does not include hemorrhoidal bleeding (DeGeorge and Nable, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:240-33).

8:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

9:

Patients with severe Crohn's disease have persistent symptoms despite outpatient medication management. Severe disease is usually characterized by high fever, frequent vomiting, evidence of intestinal obstruction, abdominal pain or tenderness, cachexia (BMI <18), or evidence of an intra-abdominal abscess, and should be hospitalized (Lichtenstein et al., Am J Gastroenterol 2018, 113: 481-517).

10:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

11:

A fistula is an abnormal communication between two internal organs or an abnormal communication which leads from an internal organ to the skin.

12:

Fistulas may occur at any location in the bowel and may be internal or external. External fistulas may be perianal (anus to skin) or enterocutaneous (bowel to skin). Internal fistulas include enteroenteric (bowel to bowel), enterovesical (bowel to bladder), enterouterine (bowel to uterus), or enterovaginal (bowel to vagina). Perianal fistulas are the most common in patients with Crohn's disease and account for 54%(0.54) of all reported fistulas; perianal fistulas may be treated in the outpatient setting. Symptoms are dependent on location which is diagnosed by radiography. Radiographic studies include: ultrasound, computed tomography (CT) scan, barium studies or fistulogram. The goal of treatment is to eradicate the fistula and reduce associated symptoms. The primary treatment includes bowel rest and parenteral nutrition; however, if an abscess is present, drainage and an anti-infective are required. Recurrence of an abscess following discontinuation of the anti-infective may require long term therapy with infliximab or azathioprine (Adegbola et al., Ann Gastroenterol 2018, 31: 129-39). When the fistula is in close proximity to the esophagus, stomach, or the proximal small intestine, it may produce a high output of gastrointestinal contents. Since this can lead to dehydration and malnutrition, monitoring fluid status is essential.

13:

Patients with fistulizing Crohn's disease may be treated with infliximab or adalimumab. If a patient does not respond to either of these tumor necrosis factor (TNF) inhibitors, experts recommend increasing the dosage or decreasing the interval between infusions and, if these strategies are not effective, switching to another TNF inhibitor (Lichtenstein et al., Am J Gastroenterol 2018, 113: 481-517).

14:

A computed tomography (CT) scan is usually the initial study in an acute intestinal fistula as it is highly specific in delineating fistulous tract anatomy. It can rule out an abdominopelvic abscess and help with surgical planning (Tuma et al., Cureus 2020, 12:).

15:

Imaging studies used in the diagnosis of ileus include abdominal computed tomography (CT) scan and abdominal x-ray.

16:

Toxic megacolon is defined as colonic dilation (greater than 6 cm) with three of the following symptoms:

- Fever
- Leukocytosis
- Tachycardia
- Anemia/bleeding

In addition to the above symptoms, the patient must have one of the following:

- Hypotension
- Hypovolemia
- Mental status change
- Electrolyte imbalance

Since hypokalemia and hypomagnesemia can exacerbate the dilation, these electrolyte imbalances should be aggressively corrected when present. Patients with toxic megacolon should be treated with intravenous hydration, broad-spectrum antibiotics, and bowel rest. Nasogastric decompression may be considered to remove air and fluid from the colon. Rapid deterioration, continued toxicity, or lack of improvement may be an indication for prompt colectomy. In most cases, surgery is performed after a short trial of medical therapy (Desai et al., Clin Exp Gastroenterol 2020, 13: 203-10; Gajendran et al., Disease-a-Month 2019:).

17:

Bowel obstruction is established by history, physical examination, and imaging studies (e.g., abdominal x-ray (KUB),

computed tomography (CT) scan, magnetic resonance imaging (MRI), ultrasonography). CT scan is the preferred imaging test for bowel obstructions as it better determines the site and cause of the obstruction (Long et al., The Journal of emergency medicine 2019, 56: 166-76). MRI may be as accurate at diagnosing bowel obstruction as CT, but is limited in its ability to identify the cause of the obstruction, has a longer scan time, and may not be readily available. Water-soluble contrast agents may provide a diagnostic as well as therapeutic benefit for small bowel obstruction. Studies of patients with partial bowel obstruction who received water-soluble contrast compared to conventional treatment alone, found there was a reduction in the need for surgery, quicker resolution, and shorter hospital stay. Additionally, a water-soluble contrast small bowel series may be helpful in determining whether conservative or operative management is appropriate (Reddy and Cappell, Current gastroenterology reports 2017, 19: 28).

18:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

19:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Semler and Rice, Clin Chest Med 2016, 37: 241-50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).

Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

20:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

21:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

22:

Enteral nutrition is delivered directly to the gastrointestinal tract and is the preferred method of feeding for patients diagnosed with inflammatory bowel disease. (Altomare et al., Nutrients 2015, 7: 2125-33). For patients who are NPO, parenteral nutrition (if necessary) should begin on day 2 for patients who present to the hospital malnourished, and on day 5 for patients who present with adequate nutritional status prior to hospitalization. This recommendation is based on clinical expert validation of best practice and available evidence.

23:

The expected treatment for most hospitalized patients with an acute flare of inflammatory bowel disease is high dose intravenous (IV) corticosteroids. In the presence of corticosteroid resistant disease or a contraindication, other medications may be used such as cyclosporine, anti-tumor necrosis factor (e.g., infliximab, adalimumab), and other immunosuppressants (e.g., azathioprine, 6-mercaptopurine, methotrexate, azathioprine, and tacrolimus). Supportive measures include deep vein thrombosis (DVT) prophylaxis, IV fluid, blood product transfusion, electrolyte replacement, and nutritional support or supplementation (Rubin et al., American Journal of Gastroenterology 2019, 114: 384-413; Lichtenstein et al., Am J Gastroenterol 2018, 113: 481-517; National Institute for Health and Clinical Excellence (NICE), NICE Clinical guideline 129.Crohn's Disease Management. 2019.).

24:

Immunosuppressant medications are used in the treatment of inflammatory bowel disease to decrease inflammation. Management of moderate to severe inflammatory bowel disease begins with the administration of intravenous corticosteroids and may progress to tumor necrosis factor inhibitors (infliximab, adalimumab, certolizumab pegol) or cyclosporine in patients who fail to respond or relapse (National Institute for Health and Clinical Excellence (NICE), NICE Clinical guideline 129.Crohn's Disease Management. 2019.; Rubin et al., American Journal of Gastroenterology 2019, 114: 384-413; Lichtenstein et al., Am J Gastroenterol 2018, 113: 481-517).

25:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

26:

Crohn's disease is a chronic inflammatory disorder of the gastrointestinal tract which is characterized by spontaneous remissions and acute exacerbations. Patients with Crohn's disease may present with abdominal pain, diarrhea, and weight loss. Extra-intestinal manifestations include: arthritis, erythema nodosum, pyoderma gangrenosum, conjunctivitis, uveitis, and renal stones. Patients with severe Crohn's disease who fail outpatient medication management may also present with a high fever, frequent vomiting, evidence of intestinal obstruction, rebound tenderness, or evidence of an intra-abdominal abscess and should be hospitalized (Lichtenstein et al., Am J Gastroenterol 2018, 113: 481-517).

27:

Parenteral nutrition is indicated in cases of small bowel obstruction where the feeding tube cannot be placed beyond the obstruction (Altomare et al., Nutrients 2015, 7: 2125-33).

28:

Inflammatory or fibrotic strictures can cause a small bowel obstruction (SBO) in uncontrolled Crohn's disease. Inflammatory strictures are more likely to respond to medical therapy such as IV corticosteroids and immunosuppressive agents than fibrotic strictures, and rarely require surgery. A patient with Crohn's disease who has had prior abdominal surgery is susceptible to a SBO, which may be an indication for surgery, rather than medical management (Reddy and Cappell, Current gastroenterology reports 2017, 19: 28).

29:

Current literature suggests patients diagnosed with bowel obstruction be made NPO and have a nasogastric (NG) tube placed to suction to decompress the dilated stomach and intestine, remove any undigested food or accumulating secretions, and prevent emesis or possible aspiration. An increase in NG tube output may indicate complete obstruction and warrants evaluation (Detz et al., Curr Probl Surg 2021, 58: 100893; Ten Broek et al., World

J Emerg Surg 2018, 13: 24).

30:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

31:

Advancing a patient's diet should occur when the patient shows signs of clinical readiness. These signs may include: a substantial improvement in pain and reduction in diarrhea, a resolution of fever if it was present at admission, absence of nausea or vomiting, and evidence of an appetite.

Advancing the diet begins with elemental feedings or clear liquids. Once tolerated, these patients progress to full liquids and ultimately to a regular diet. With the initiation of a regular diet, the patient may resume their oral medications.

32:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

33:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360:

j5790).

34:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

35:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

36:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

37:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

38:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

39:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

40:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives. Guidelines describe multiple acceptable regimens but do indicate a preference for oral fidaxomicin for patients with mild to moderate disease. Alternative regimens are driven by disease severity and the number of episodes the patient has experienced. Treatment guidelines recommend discontinuation of the inciting anti-infective as soon as possible to prevent recurrence. Other interventions include IV fluids, electrolytes, or oral fluids to maintain hydration. Use of opiates and antidiarrheal medications should be avoided because decreased intestinal motility will exacerbate the infection (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

41:

Instruction: This criteria point is intended for anti-infectives initiated for the treatment of *C. difficile* infections. The preferred treatment of *Clostridioides difficile* infections is based on the episode of illness and the severity of the symptoms. The Infectious Diseases Society of America (IDSA) recommends a variety of treatment regimens which are dependent on clinical presentation, including disease severity and how many episodes of disease the patient has experienced (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

42:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit (Scales, Br J Nurs 2014, 23).

43:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

44:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

45:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

46:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status;** those who

do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

47:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

48:

Signs indicating recovery from *Clostridioides difficile* include:

- Decrease in the severity and frequency of diarrhea
- Absence of fever
- Normal white blood cell count

Not all patients demonstrate fever or leukocytosis during the course of illness. Diet advancement should be based on improving clinical signs and symptoms.

49:

Fistula management for patients with Crohn's disease consists of the application and maintenance of the drainage system, protection of skin around the fistula to prevent breakdown, and management and evaluation of the fistula. Another aspect includes teaching the patient or caregiver to perform these functions.

50:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

51:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

52:

Inflammatory bowel disease (IBD) is a complex, chronic disease that may be challenging to manage for both patients and caregivers. Risk for readmission in this population is higher in those with psychiatric disease, tobacco use, chronic pain, or noncompliance with outpatient follow-up. Those at high risk for readmission should be identified for improved coordination of care to decrease readmission and improve patient outcomes (Barnes et al., Inflammatory bowel diseases 2017, 23: 875-81).

53:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

54:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to

identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

55:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

56:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

57:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

58:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs.

Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

59:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

60:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

61:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

62:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal

relationships). Pain relievers should also be included

- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

63:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

64:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

65:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

66:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require

total assistance for functional mobility or ADLs

67:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

68:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

69:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

70:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

71:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.